Pehchaan for Paytm merchants

Never lose a *regular* again.

Customer intelligence for every kirana — built from the UPI payments Paytm merchants already receive.

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Track 1 — Merchant Growth AI + an autonomous Track 3 agent

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pehchaan-sigma.vercel.app

THE PROBLEM

Quick commerce is taking the kirana's best customers

51%

of shoppers say they rely less on their kirana than a year ago.

Grant Thornton Bharat survey, via NEXT IAS (Jul 2026)

They leave silently

Nobody cancels a kirana. They order from an app one week and simply don't come back.

400 customers, zero visibility

A shopkeeper can't notice that a regular who came every 3 days hasn't come in 3 weeks.

THE INSIGHT

Every UPI payment already carries a *name.*

Paytm sees the payer name and UPI ID on every payment across **~1.5 crore subscription merchants** — and today uses it only to announce “payment received”.

Subscription merchants at end of FY26 — Inc42

THE SOLUTION

Pehchaan: from a payment to a relationship

01

Recognise

Each Soundbox payment builds a customer profile automatically: how often, how much, which household.

02

Predict

A probabilistic churn model scores every customer's chance of still being a regular — daily.

03

Act

An agent picks who's worth winning back, writes to them in Kannada, and confirms their return at the counter.

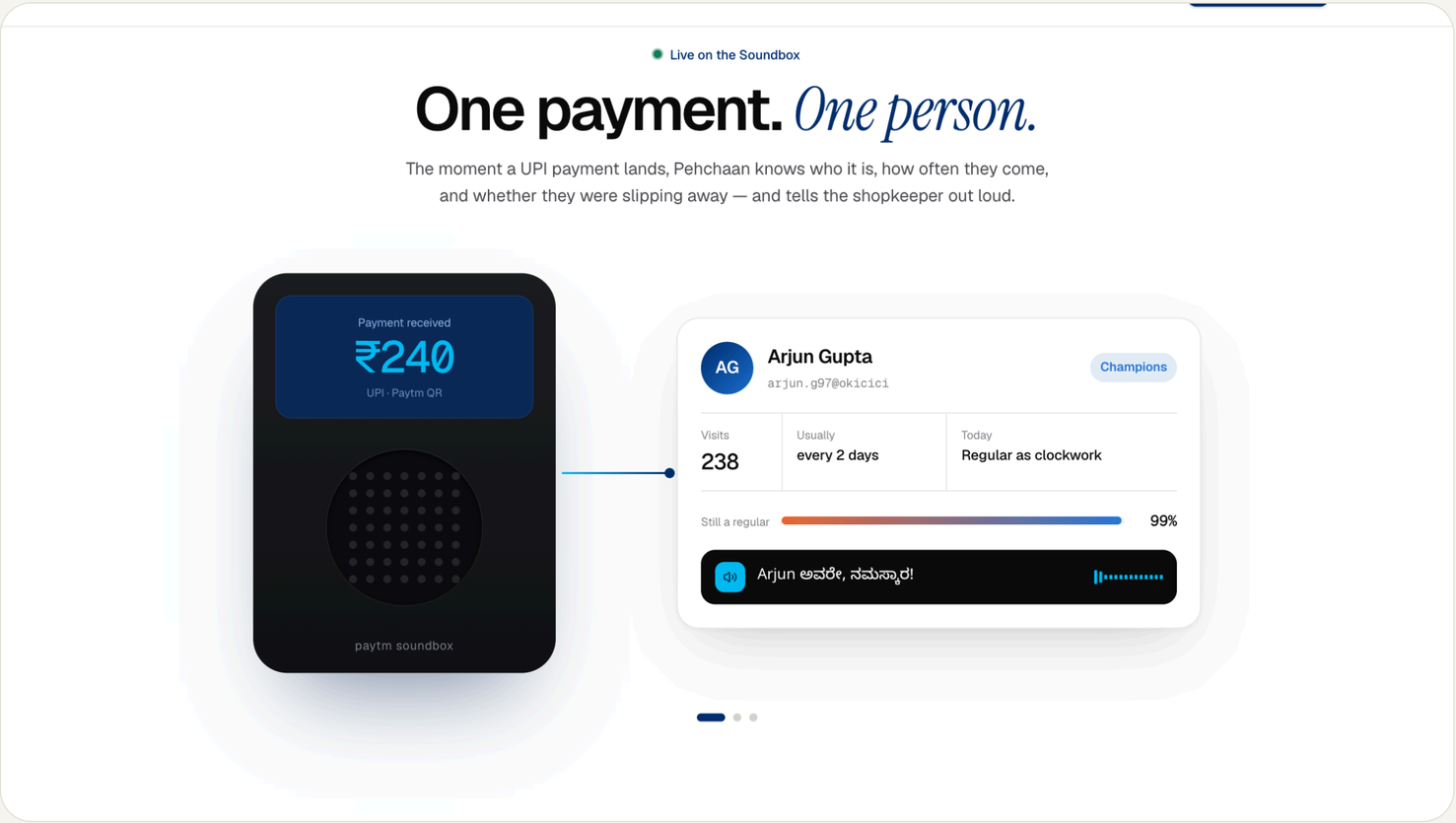
04

See the city

Across shops, Paytm spots regulars leaving the same week — and locates the new competitor.

PRODUCT · AT THE COUNTER

One payment. One person.

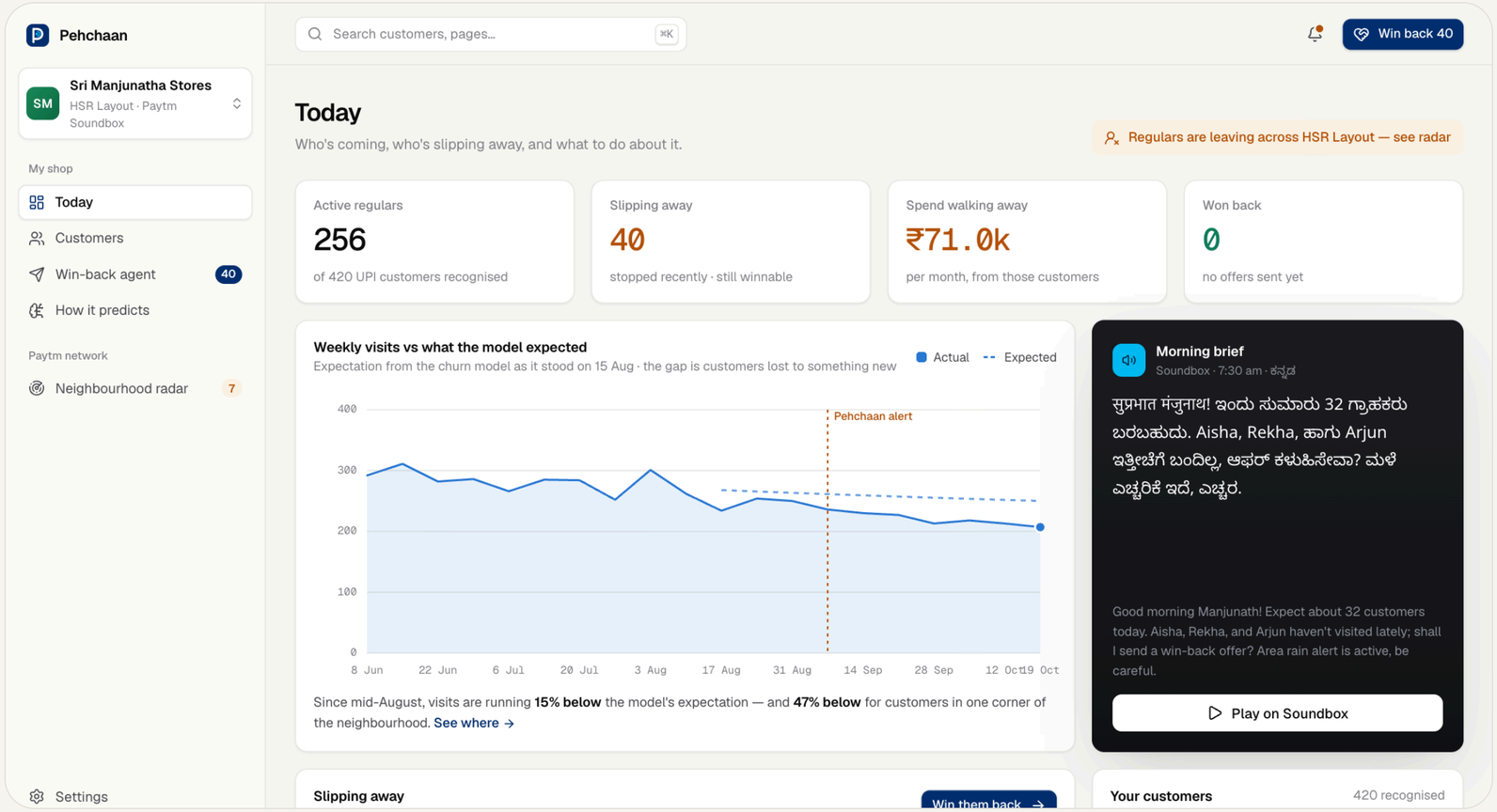


When ₹240 lands, the Soundbox knows it's **Arjun** — visit 238, every 2 days, 99% still a regular — and greets him by name.

When a lapsed regular returns, it tells the owner: “ಆಯಿಷಾ ಅವರು 48 ದಿನಗಳ ನಂತರ ಮತ್ತೆ ಬಂದಿದ್ದಾರೆ!”

PRODUCT · MERCHANT DASHBOARD

Today: who's slipping away, and what to do



- Visits vs what the model expected — the gap is customers lost
- Morning brief, spoken in Kannada on the Soundbox
- Slipping-away list, ranked by monthly spend × churn probability

HOW IT PREDICTS

Real machine learning — not an LLM wrapper

Payments

Per UPI ID: frequency, recency, customer age, spend

→

BG/NBD

P(still a regular) and expected visits — Fader, Hardie & Lee 2005

→

Gamma-Gamma

Spend per visit → 90-day customer value

→

Priority

Who to win back: monthly value × churn probability

Built for silent churn

A daily regular gone 10 days matters; a monthly one gone 10 days doesn't. The model knows the difference — a fixed “30 days” rule can't.

Where the LLM fits

Gemini only writes the Kannada messages and the morning brief. Sarvam speaks them. Neither decides who or how much.

PROOF

Backtested on months the model never saw

2,891

visits predicted

for 381 customers, 1 Jun – 15 Aug, trained only on data before 1 Jun

2,899

visits that happened

−0.3% total error

0.92

per-customer correlation

mean absolute error 2.5 visits

41%

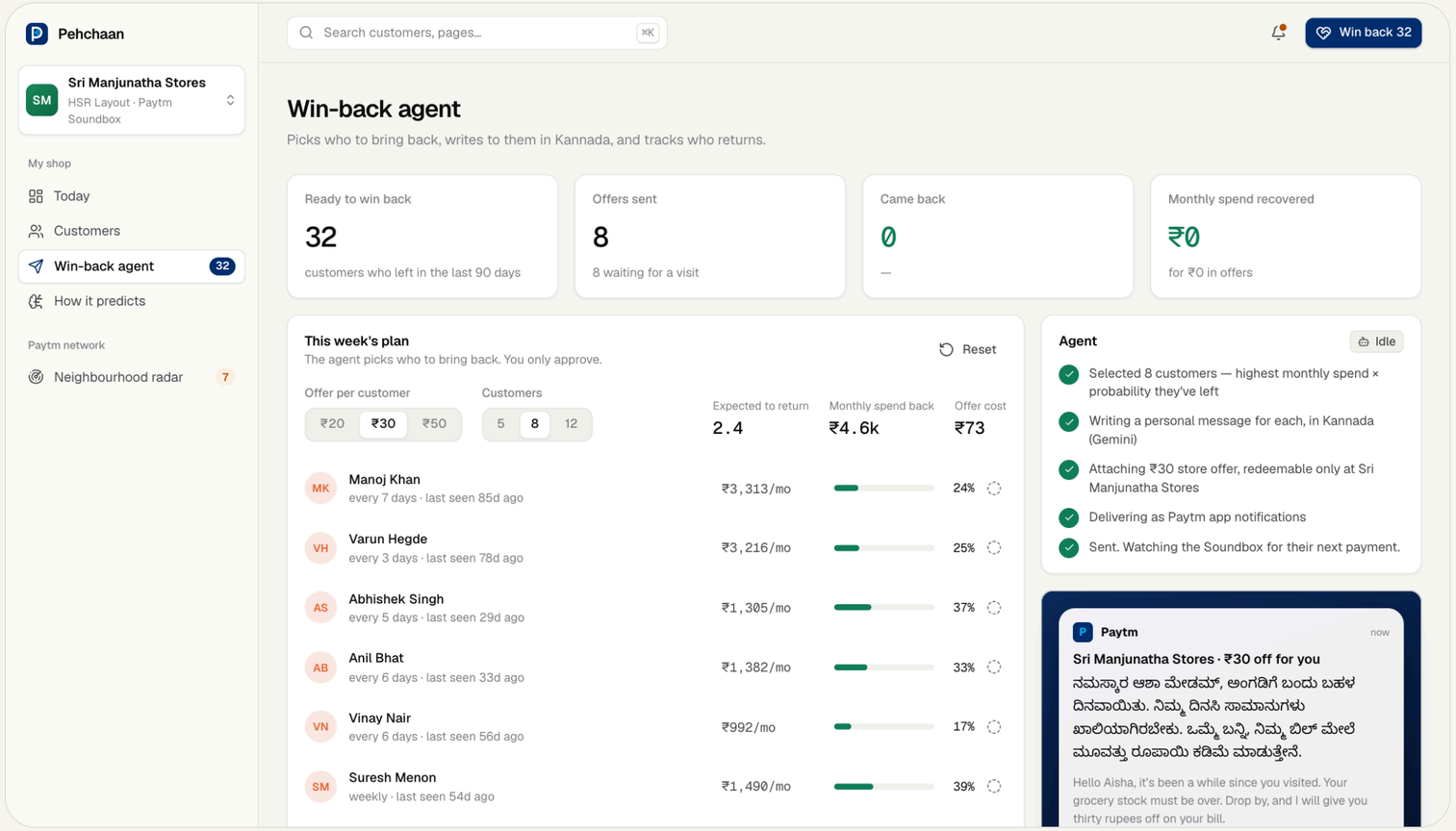
of all visits

came from the model's top 10% of customers

Demo data is a deterministic simulation of one HSR Layout kirana — 420 customers, 14,127 payments. The models are unchanged for real Paytm payment histories; a pilot replaces the simulation. Reproduce it: github.com/SayantikaLaskar/pehchaan → `npm run model:check`

TRACK 3 · WIN-BACK AGENT

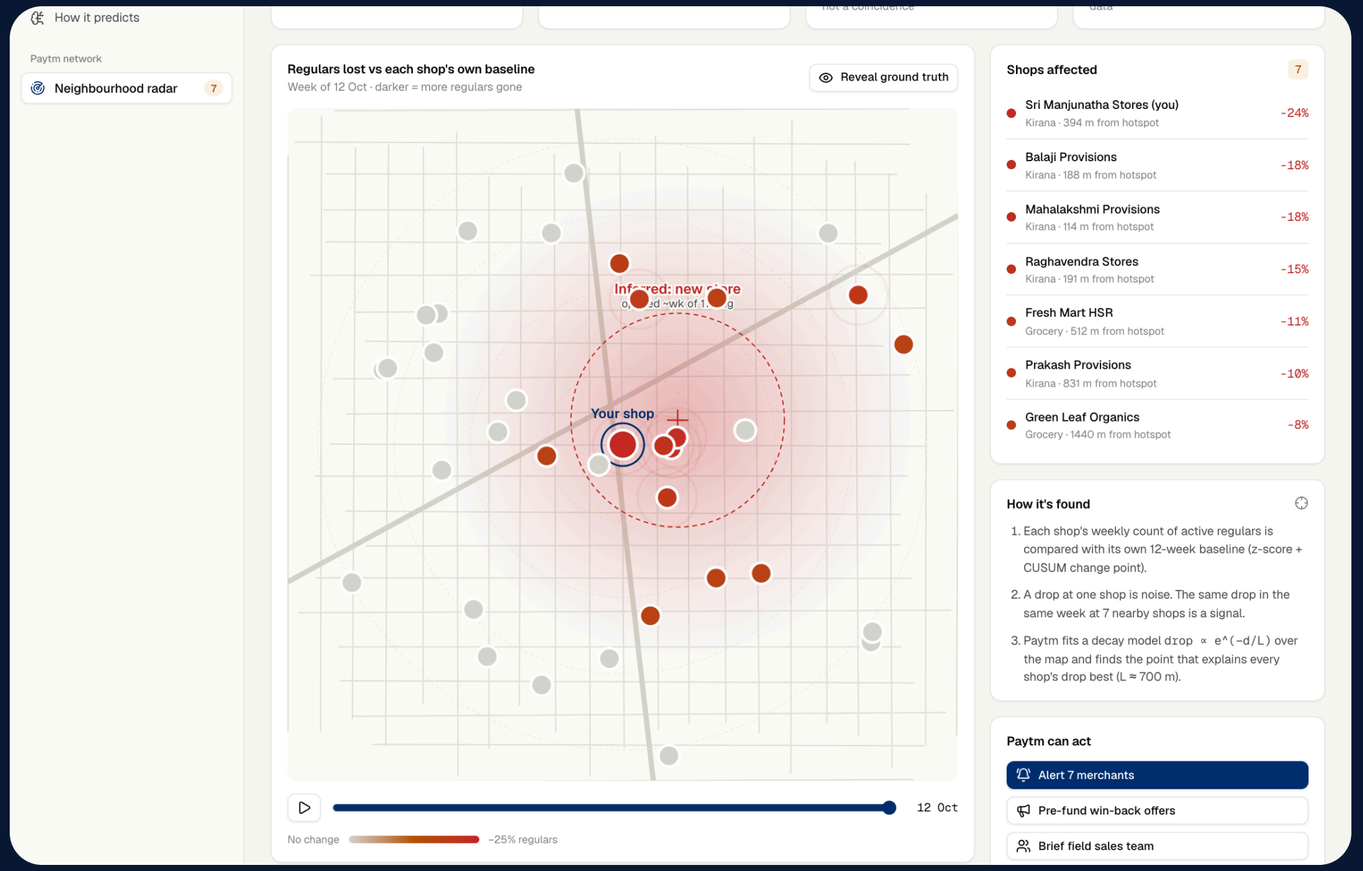
An agent that gets the job done



1. **Decides** who's worth an offer — and who isn't
2. **Writes** a personal note in Kannada
3. **Sends** it as a Paytm app notification
4. **Verifies** the return when they pay at the Soundbox

NEIGHBOURHOOD RADAR · PAYTM VIEW

Paytm sees what no single shop can



7 of 36
shops lost regulars in the same weeks

wk of 17 Aug
change point found — store opened 18 Aug

~360 m
error locating it, from aggregate drops alone

TECHNOLOGY

Tech stack

Layer	What we use
Data	Paytm QR / Soundbox payments — payer name + UPI ID, already collected
Models	BG/NBD, Gamma-Gamma, CUSUM change points, spatial decay fit — TypeScript, maximum-likelihood
Agent	Gemini Flash-Lite — structured JSON output, cached, with templates as fallback
Voice	Sarvam AI — Bulbul v3 Kannada speech, Saaras v4 recognition
App	Next.js 15, React 19, GSAP motion — merchant dashboard + Paytm network view
Channels	Soundbox voice at the counter · Paytm app notifications to customers

USP

Why Pehchaan is different

Capability	Billing / loyalty apps	Generic AI chatbot	Pehchaan
Works with zero data entry	No	No	Yes
Predicts who is leaving	No	No	Yes
Acts on its own (win-back)	No	No	Yes
Speaks Kannada at the counter	No	Yes	Yes
Sees neighbourhood-level shifts	No	No	Yes

Our moat is Paytm's: the Soundbox at the counter, the consumer app to reach customers, and payment data across shops.

IMPACT

Impact & benefits

MERCHANT

₹71k / month

- of spend slipping away made visible at one kirana (simulation)
- Win back regulars before they're gone
- No app, no typing — the Soundbox speaks

CUSTOMER

By name

- Recognised and greeted at the counter
- Offers that fit what they actually buy
- Delivered under Paytm consent

PAYTM

1.5 Cr merchants

- Soundbox retention vs rival devices
- More consumer-app opens via store offers
- Radar for sales and partnerships

BUSINESS MODEL

Free to know your customers. Paid to win them back.

Pehchaan

Free

bundled with every Soundbox

- Customers recognised
- Slipping-away alerts
- Kannada morning brief

Pehchaan Pro

₹99

per month (proposed)

- Autonomous win-back agent
- Offers via the Paytm app
- Weekly ₹-recovered report

Paytm network

Take rate

on merchant-funded offers

- Brand-funded kirana offers
- Radar for sales teams
- Device retention

One regular won back (~₹2,000 / month in spend) pays for a year of Pro.

ROADMAP

Risks, mitigations and an 8-week pilot

Risk	Mitigation
Privacy (DPDP Act)	Merchants never see phone numbers; offers go via Paytm app consent with opt-out; network views are aggregated across shops.
Cash customers	Only UPI regulars are scored — most repeat customers in a city kirana, and the share is growing.
Shared phones & UPI IDs	Household grouping links IDs by surname, timing and basket; the owner can merge or split profiles.
Offer fatigue	Champions never get discounts; one offer per customer per month; agent learns from redemptions.

Pilot · 50 kiranas, HSR Layout

Wk 1-2 — **Onboard & fit**
Soundbox opt-in, 6 months of payment history scored

Wk 3-6 — **Agent live**
Weekly win-backs; 20% of at-risk customers held out as control

Wk 7-8 — **Measure**
Return rate vs control, ₹ recovered per shop, merchant retention

Success: return rate ≥ 2× control at positive ROI

Never lose a *regular* again.

Live demo · pehchaan-sigma.vercel.app
Code · github.com/SayantikaLaskar/pehchaan

Sayantika Laskar & Harsh Shukla
Thank you